EECE2140 - Final Project Iteration #2
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DEFINED TASKS

Tasks	Task ID	Menu Opt.	Assignee	File	Function
Configure Pack	T1	1	Kavi	model_tasks.cpp /	configurePack()
Show thermal state	T2	2	Kavi	model_tasks.cpp /	showThermal()
Run Charge Cycle	T3	3	Drew	control_tasks.cpp/	chargeCycle()
Run Discharge (CC/pulsed/CSV)	T4	4	Anthony	sim_tasks.cpp/	dischargeCycle()
Show SOC estimate vs. true	T5	5	Drew	control_tasks.cpp/	showSOC()
Show energy report	T6	6	Drew	control_tasks.cpp/	reportEnergy()
Set thresholds for protection	T7	7	Drew	control_tasks.cpp/	setThresholds()
View event log	T8	8	Anthony	sim_tasks.cpp/	viewEvents()
Toggle CSV log	T9	9	Anthony	sim_tasks.cpp/	toggleLog()
Exit	T0	0	Anthony		case 0

EXPECTED DELIVERABLES/STATUS

- **Configure Pack:** Set cell count, capacity, and resistance **[STUB]**
- **Show thermal state:** Perform equations for temperature of each cell **[STUB]**
- **Run Charge Cycle:** Hold constant current until 4.2V is met, at which point constant voltage is held; live status and stop on fault **[STUB]**
- **Run Discharge (CC/pulsed/CSV):** Allow user to choose between constant-current, pulsed, or CSV load profile, then discharge pack until low SOC, undervoltage, or fault ends run **[STUB]**
- **Show SOC estimate vs. true:** display counted SOC next to true SOC along with estimation error **[STUB]**
- **Show energy report:** display Wh_{in} and Wh_{ou}, trip efficiency, and cycle count **[STUB]**
- **Set thresholds for protection:** Allow user to edit WARNING/FAULT levels for over/undervoltage, overcurrent, overtemperature, low SOC, and reset faults **[STUB]**

- **View event log:** Print timestamps for warnings, faults, and charge/discharge **[STUB]**
- **Toggle CSV log:** Turn CSV logging on/off and set output file name; each row records time, V, I, SOC, temperature, and fault state **[STUB]**
- **Exit:** End loop and close all open log files before exiting program **[STUB]**

SYSTEM DESIGN

Class	Owner	Private Data	Public Methods
Cell	Kavi	capacity (Ah), SOC, base internal, resistance, temperature, present current, OcvTable, ThermalModel	update(current, dt), getTerminalVoltage(), getOCV(), getResistance(), getSOC(), getTemperature()
OcvTable	Kavi	SOC points/matching OCV values	lookup(soc), getMinVoltage(), getMaxVoltage()
ThermalModel	Kavi	cooling coefficient k, thermal capacity Cth, ambient temperature	step(temperature, current, resistance, dt), setAmbient(temp), getAmbient()
Pack	Kavi	cells, number of cells in series	configure(numCells, capacity, resistance, startSOC), applyCurrent(current, dt), getVoltage(), getMaxCellVoltage(), getMinCellVoltage(), getMaxTemperature(), getSOC(), printCellStatus()
Charger	Drew	CC current, CV voltage, cutoff current, mode	getChargeCurrent(pack), getMode(), isDone(), reset()
SocEstimator	Drew	Estimated SOC, capacity (ah), noise level, sensor offset, random number generator	update(trueCurrent, dt), getEstimate(), getError(trueSOC), reset(startSOC)
EnergyTracker	Drew	Wh_{in}, Wh_{out}, cycle count, charge throughput	update(voltage, current, dt), getWhIn(), getWhOut(), getEfficiency(), getCycleCount(), reset()

Protection	Drew	warning and fault limits for each check, latched-fault flag, active warnings	check(pack, current), setLimit(check, level, value), isFaulted(), resetFault(), getActiveWarnings()
Contactor	Drew	open/closed state	open(), close(), isClosed()
LoadProfile	Anthony	profile types, constant current, pulse high/low, current and period, CSV time and current points	setConstant(current), setPulsed(high, low, period), loadCSV(filename), getCurrent(time)
Logger	Anthony	output file stream, enabled flag, file name	enable(filename), disable(), isEnabled(), logRow(time, V, I, SOC, estimated SOC, temperature, fault state)
EventLog	Anthony	list of timestamped messages	add(time, message), print(), clear()
Simulator	Anthony	one of each object above, simulation time, timestep	runCharge(), runDischarge(), step(current), printStatusLine(), plus getPack(), getProtection(), and similar accessors for the task functions

MENU

BMS SIMULATOR:

1. Configure pack
2. Show thermal state
3. Run charge cycle
4. Run discharge (CC/pulsed/CSV)
5. Show SOC estimate vs. true
6. Show energy report
7. Set protection thresholds
8. View event log
9. Toggle CSV logging
0. Exit

Select an option:

FILE STRUCTURE

bms/
|_ Makefile
|_ README.md
|_ .gitignore
|_ docs/
|_ data/
|_ include/
| |_ tasks.h
| |_ cell.h ocv_table.h thermal_mode.h pack.h
| |_ charger.h soc_estimator.h energy_tracker.h protection.h contactor.h
| |_ load_profile.h logger.h event_log.h simulator.h
|_ src/
| |_ main.cpp
| |_ model_tasks.cpp [T1-T2 (Kavi)]
| |_ control_tasks.cpp [T3, T5-T7 (Drew)]
| |_ sim_tasks.cpp [T4, T8-T9 (Anthony)]
| |_ one .cp per class, matching include/
| |_
|_ tests/